EXERCISE 2

Superposition for Homogeneous Linear Systems

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Superposition • Linear Combination of Solutions

STATEMENT

Show that if a homogeneous linear system

$$\begin{cases} \dfrac{dx}{dt} = a_1(t)x + b_1(t)y, \\ \dfrac{dy}{dt} = a_2(t)x + b_2(t)y \end{cases} \tag{1}$$

has two solutions

$$\begin{cases} x = x_1(t) \\ y = y_1(t) \end{cases} \quad \text{and} \quad \begin{cases} x = x_2(t) \\ y = y_2(t) \end{cases} \tag{2}$$

then any linear combination

$$\begin{cases} x = c_1x_1(t) + c_2x_2(t) \\ y = c_1y_1(t) + c_2y_2(t) \end{cases} \tag{3}$$

with constants $c_1, c_2 \in \mathbb{R}$ is a solution as well.

PHASE 1: SETUP

Hypotheses: a_i, b_i are functions of t (not necessarily constant) defined on an interval I, and (x_1, y_1), (x_2, y_2) are differentiable on I with

$$\dot{x}_j = a_1 x_j + b_1 y_j, \qquad \dot{y}_j = a_2 x_j + b_2 y_j \qquad (j = 1, 2) \tag{H}$$

PHASE 2: FORMULATION

To show: with $x = c_1 x_1 + c_2 x_2$ and $y = c_1 y_1 + c_2 y_2$ the two equations of (1) hold. Tool: differentiation is linear, $\frac{d}{dt}(c_1 f + c_2 g) = c_1 \dot{f} + c_2 \dot{g}$ for constants c_1, c_2. In vector notation: $\mathbf{x} = c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2$ solves $\dot{\mathbf{x}} = A(t)\mathbf{x}$ because the right-hand side is linear in $\mathbf{x}$ (Robinson Eq. 27.3).

PHASE 3: DERIVATION

First equation. Differentiate $x = c_1 x_1 + c_2 x_2$ and use (H):

$$\dot{x} = c_1 \dot{x}_1 + c_2 \dot{x}_2 = c_1 (a_1 x_1 + b_1 y_1) + c_2 (a_1 x_2 + b_1 y_2)$$

Regroup the terms with a_1 and with b_1:

$$\dot{x} = a_1 (c_1 x_1 + c_2 x_2) + b_1 (c_1 y_1 + c_2 y_2) = a_1\, x + b_1\, y$$

Second equation. In the same way

$$\dot{y} = c_1 \dot{y}_1 + c_2 \dot{y}_2 = c_1 (a_2 x_1 + b_2 y_1) + c_2 (a_2 x_2 + b_2 y_2) = a_2 (c_1 x_1 + c_2 x_2) + b_2 (c_1 y_1 + c_2 y_2) = a_2\, x + b_2\, y$$

Both equations of (1) are satisfied by (3). ■

Matrix form

$$\dot{\mathbf{x}} = c_1 \dot{\mathbf{x}}_1 + c_2 \dot{\mathbf{x}}_2 = c_1 A \mathbf{x}_1 + c_2 A \mathbf{x}_2 = A(c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2) = A\mathbf{x}, \qquad A(t) = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$$

PHASE 4: INTERPRETATION AND CHECKS

- Linearity and homogeneity are both used: homogeneity guarantees that the right-hand side of (1) vanishes when $x = y = 0$ (a forcing term $f(t)$ would be multiplied by $c_1 + c_2$ instead of by 1, and the combination would not be a solution unless $c_1 + c_2 = 1$ and so on).
- Constant coefficients were not needed: only linearity of the derivative and of $(x, y) \mapsto (a_1 x + b_1 y, a_2 x + b_2 y)$.
- The set of solutions is therefore a vector space; Exercises 3 and 4 show that it has dimension 2.
- Quick example: $\dot{x} = y, \dot{y} = -x$ has $(\cos t, -\sin t)$ and $(\sin t, \cos t)$ as solutions, and so $(c_1 \cos t + c_2 \sin t,\ -c_1 \sin t + c_2 \cos t)$ does.